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Biocurrent regulates the mercury transformation in paddy soil

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ABSTRACT

Mercury (Hg) pollution in farmland threatens the soil health and agricultural product safety. Bioremediation is an environmentally friendly approach, and yet the effects of soil microbial electron transfer on Hg conversion remain unclear. Thus, a 135-day soil microbial electrochemical system (MES) was established to investigate how Hg forms are regulated by directionally accelerated electron transport. The results revealed that the flux of soil electron transfer was determined by the transformation of Fe(III)/Fe(II). The bioavailable Hg and methylmercury (MeHg) contents were reduced by 16–29% and 37−45%, respectively, in the soil MES on Days 45 and 90 compared with those in the control, whereas the total Hg content decreased by 4.6–7.6% compared with that in the original soil (OS) on Day 135 ($p < 0.05$). Biocurrent regulates Hg transformation by stimulating microorganisms carrying Hg transformation genes. Compared with those in OS, the abundances of *Geobacter* and *Clostridium*, which are responsible for bioelectricity generation and MeHg conversion, in the anode increased by 42-fold and 99%, respectively. Interestingly, the upregulation of the *merA/B* gene increased MeHg demethylation, although the number of *hgcAB* gene copies also increased. The results of the plant experiments revealed that the treated soils had no adverse effects on rape growth and Hg accumulation. We proposed a biogeochemical model for the regulation of Hg transformation by biocurrent driven by the iron reduction process in paddy soil, which provides a supporting for predicting and controlling the production of MeHg in paddy soil. This study highlights the impact of biocurrent on Hg transformation in soil and offering a new approach for the safe management of Hg-contaminated soil.

1. Introduction

Mercury (Hg) pollution results from natural processes (e.g., volcanic activity and biomass burning) and human activities (e.g., fossil fuel combustion and nonferrous metal mining), with soil being the largest reservoir (Driscoll et al., 2013). Hg in soil can exist in various forms and can cause different types of harm to humans (Schartup et al., 2015). Such as, Hg can cause damage to the human nervous system, damage to the reproductive system, weaken the immune system and so on (Kim et al., 2016). The common valence states of Hg are Hg^0, Hg^+ and Hg(II), among which Hg^+ is unstable and prone to disproportionation reactions into Hg^0 and Hg^{2+} (Liem-Nguyen et al., 2021). Hg(II) has the highest bioavailability in paddy soil and easily produces methylmercury (MeHg) with high biological toxicity (Zhou et al., 2022). Rice fields are hot spots of MeHg formation; therefore, rice is a main source of MeHg intake by humans (Feng et al., 2008; Hao et al., 2022). MeHg, the most toxic organic Hg form, can enter human blood and follow the blood circulation to poison tissues, mainly the liver, kidney and brain (Beckers and Rinklebe, 2017; Liu et al., 2023). Therefore, the amendment of Hg-contaminated rice fields is urgently needed.

MeHg is produced mainly by anaerobic microorganisms (C.-J. Zhang et al., 2023), e.g., *Proteus deltaic*, *Firmicutes* and methylated methane microorganisms, whereas abiotic processes are negligible (Jones et al., 2019). Hg-methylating microorganisms include mainly iron-reducing

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bacteria (IRB) (Yu et al., 2012), sulfate-reducing bacteria (SRB) (Jones et al., 2019) and methanogens (Brauner and Briggs, 2023; C. Chen et al., 2019). IRB can use Fe(II) to provide electrons and Fe(III) as an electron acceptor while carrying out the methylation of Hg; for example, *Geobacteraceae* is a key predictor of MeHg bioaccumulation in paddy soils (Merino et al., 2021; Zhong et al., 2024). During Hg methylation by SRB, the use of sulfate (SO_4^{2-}) as the main electron acceptor improves *hgcA* abundance and thus Hg methylation (Liu et al., 2023). Methanogenic bacteria can use carbon dioxide and acetic acid as substrates for growth and methylation (Friedrich, 2005). Redox reactions of the above-mentioned processes depend on exogenous electron donors. Accelerating soil electron transport can increase the number of electron donors required by bacteria and promote energy metabolism (Ouboter et al., 2024). In addition, some microorganisms contain mer operons, which play a key role on Hg cycle. The *merA* gene reduces Hg(II) to Hg^0, and the *merB* gene encodes the demethylation of MeHg (Lu et al., 2016). However, the effects of accelerated soil electron transport on Hg methylation remain unclear.

Microbial Hg methylation pathways include the acetyl-CoA (Wood–Ljungdahl) pathway, methionine cycle pathway and methanogenesis pathway (S. C. Choi, T. Chase, & R. Bartha, 1994a; S. C. Choi, T. Chase, Jr., & R. Bartha, 1994b). In 2013, Parks cracked the Hg methylation gene code (*hgcAB*), which was identified in all microorganisms with the ability to methylate Hg (Parks et al., 2013). HgcA and HgcB are *hgcAB* translational proteins that are responsible for methyl and electron transfer during methylation (Capo et al., 2022). The *hgcB*-bound [4Fe-4S] protein is the simplest electron transfer group found in biological systems and can accept electrons from certain substrates to reduce Co(III) in the corrin ring to the more active Co(I) (Podar et al., 2015). Electron transfer can alter microorganisms' ability to remove methyl groups by affecting their metabolic pathways. For example, it may increase their affinity for electron acceptors, thereby promoting the activity of specific metabolic pathways (Y. Wang et al., 2025). Additionally, environmental factors such as temperature, pH, and nutrient availability can influence the rate and efficiency of electron transfer, thereby impacting the demethylation capacity of microorganisms (Guan et al., 2024). Obviously, electron transfer is an important step in microbial methylation and demethylation.

A soil microbial electrochemical system (MES) provides an inexhaustive electron acceptor and donor, which can accelerate the transfer of electrons generated by soil microbial decomposition of organic matter (H. Wang et al., 2024; K. Wang et al., 2023). This study hypothesizes that accelerating electron transport in Hg-contaminated soils via MES will alter mercury bioavailability and methylation by influencing microbial community structure and function. By altering the chemical form of Hg, microorganisms not only influence its environmental distribution and bioavailability but also significantly affect its toxicity and ecological impact (Ruuskanen et al., 2020). Therefore, understanding the role of these microorganisms in Hg transformation is crucial for predicting and controlling Hg pollution. In this study, we found that biocurrent can reduce the bioavailability of Hg. Biocurrent promoted the enrichment of functional microorganisms and stimulated the expression of Hg conversion genes *merA* and *merB*, accelerating the Hg cycling process. Our results offer new insights into the safe management of Hg-contaminated paddy soil by enhancing iron reduction through accelerated electron transfer, which in turn speeds up demethylation and atmospheric Hg circulation, ultimately helping to reduce Hg pollution.

2. Materials and methods

2.1. Sample collection

Soil was taken near the Wanshan Hg mine in Tongren city, Guizhou Province. Two kinds of soils marked as A (umber) and B (red soil) with different Hg contents were taken from Tianchong village (109°05′14″N, 27°37′42″E) and Shaping village (109°12′13″N, 27°46′06″E), respectively. The soil samples were naturally air-dried in the greenhouse, with stones and plant and residues removed. The samples were then sieved through a 2 mm sieve (10 mesh), thoroughly stirred, and subsampled using the quartering method. All procedures were completed within 3 days to minimize Hg volatilization and contamination from exogenous Hg-containing substances. The samples were stored in sealed bags, kept in the dark, and stored at 4 °C for subsequent analysis. The soil properties (pH, EC, OM, DOC, TC, SO_4^{2-} and HCl-extractable Fe) were determined according to conventional methods (Texts S1). The physical and chemical properties of the two soils are shown in Table S1.

2.2. Soil MES construction and operation

The reactor was a polypropylene plastic box (10 × 10 × 9.5 cm) with a titanium sheet as the wire and carbon felt as the anode and cathode. The sizes of the anode and cathode were 6 × 6 cm and 8 × 8 cm, respectively. Mix thoroughly 500 g of soil with 150 mL of water, pour part of the soil into the reactor and put the anode when the height reaches 3 cm, then add the remaining soil, put the cathode (surface layer), and add water (3 cm in height) to seal (Fig. S1). Water was replanned to the same height daily using a syringe during incubation in the reactor. The closed-circuit groups and nonelectrode control were denoted CC and CK, respectively. When the signal acquisition system was connected, the voltage (U, V) of the external resistance (R, 1000 Ω) was recorded every minute (Zhao et al., 2019). Some reactors were dismantled on Days 45, 90 and 135 to gain soil samples. In order to better investigate the effect of biocurrent on Hg conversion, we set the sampling points near the cathode (Cat) and anode (Ano). Soil MES device design and sampling points are displayed in Fig. S1. All soil samples were stored at −20 °C.

2.3. Analytical methods for electrochemistry

The extracellular polymeric substances (EPSs) in soil were determined by cation exchange resins (CERs) (Redmile-Gordon et al., 2014). The 0.01mol $CaCl_2$ was used as the extractant, and neutral phosphate buffer was used to maintain the stability of the soil biological cells. CER was used to accelerate the dissolution of EPSs. The protein components (Pro) in EPSs were determined via double myosin acid, the polysaccharide components (Pol) were determined via the phenol–sulfuric acid method, and the content of cytochrome *c* (Cyt *c*) was determined via a Cyt *c* enzyme-linked immunosorbent assay kit.

The charge output was calculated by $Q = \int_0^T (U/R)\mathrm{d}t$, where U is the collected voltage, R is the external resistance, T is the running time, and the integral is 1800 s as a unit gradient (X. Chen et al., 2021; Yang et al., 2024). The power density curve and polarization curve of the reactor were tested by gradually reducing the external resistance. Before the test, the open-circuit situation was maintained for 24 h to ensure system stability. The resistance was reduced step by step from 10000 to 100 Ω, and the output voltage was measured after each resistance value was stabilized for 25 min. The biocurrent density (I, $mA{\cdot}m^{-2}$) was calculated via $I=U/(R{\cdot}A)$, where A is the effective cathodic area. The power density was calculated as $P=U{\cdot}I$ (Han et al., 2022).

2.4. Hg form analysis

The soil THg content was determined according to the method described in GB/T 22105.1–2008. A sample of 0.1 g air-dried soil through a 100-mesh nylon mesh screen was placed in a polypropylene digestion tube, digested by 1 + 1 (Nitric acid + hydrochloric acid), and determined by atomic fluorescence spectrophotometer AFS8520 (Haiguang, China). The standard material GBW07311 and the soil standard reference material SRM 2587 (NIST) were used for quality control (recovery 85.2%–101.7%, relative deviation (RSD) < 9%). The detection

limit of the method was 0.01 μg kg^{-1} (Pei et al., 2022; Yin et al., 2022).

The soil Hg forms were extracted via the Tessier continuous extraction method (Pei et al., 2022), which included the exchangeable state (EX), carbonate binding state (CB), Fe–Mn oxide binding state (OX), organic matter binding state (OM) and residue state (RS). The test methods are described in Texts S1.

The bioavailable Hg (Bio Hg) in the soil was extracted with 0.01 mol L^{-1} $Na_2S_2O_3$ as a one-step chemical reagent. The ratio of *m* (soil) to *v* (water) was 1:10, and the mixture was oscillated for 4 h. The validity of the Hg was determined via an atomic fluorescence photometer (AFS 8520) after centrifugation (Huang et al., 2020).

MeHg in the soil was determined via HNO_3 and $CuSO_4$ solution extraction and CH_2Cl_2 extraction combined with water-phase ethylation (GC-CVAFS). The average recovery of the method was 97.8%, the relative standard deviation was less than 10.2%, and the detection limit was 0.6 pg·g^{-1} (Liu et al., 2023).

2.5. Microbial analysis and functional genes

A kit (Tiangen, Germany) was used to extract the soil DNA. The primers 515F (GTGCCAGCMGCCGCGGTAA) and 806R (GGACTACHVGGGTWTCTAAT) were used to amplify the V4 region of 16S rRNA (Yang et al., 2024). The PCR products were detected via electrophoresis via 2% agarose gel. Genomic libraries were constructed via the NEBNext Ultra™ II FS DNA PCR-free Library Prep Kit (New England Biolabs), quantified via Qubit and qPCR and finally sequenced on a NovaSeq 6000 instrument (Illumina, US) by Novogene (Beijing, China).

In addition, we used absolute quantitative PCR to detect Hg-transforming genes, including the methylation gene *hgcAB*, the Hg reduction gene *merA*, and the MeHg reduction gene *merB*. Furthermore, since the methylating microorganisms in the previous study were primarily IRB and SRB, we also identified the representative genes for IRB (*FOB*) and SRB (*dsrAB*) and analyzed by Guangdong Meg Gene Technology Co., Ltd. Nonspecific fluorescent markers were used to calculate the number of gene copies. The real-time fluorescent quantitative PCR test steps and gene primers are shown in Texts S2 and Table S2, respectively.

2.6. Effects of MES on plant growth and Hg accumulation

In order to explore the effects of soil MES treatment on plant growth and Hg accumulation, a pot experiment was carried out. Rape (Brassica napus) has a short growth cycle and is often used as a model crop for toxicity verification. We conducted pot experiments using both closed-circuit treated soil and nonelectrode treated soil. Both kinds of soils were tested to better confirm the safety by a rapeseed cultivation experiment. The first batch of experiments (using soil on Day 45) was sown in November 2023 and harvested after 50 days of culture. The second batch of experiments (using soil on Days 90 and 135) was sown in April 2024 and harvested after 50 days of culture. All pot experiments were conducted in an intelligent greenhouse, with the room temperature maintained at an average of 15–20 °C. To prevent the potential effect of Hg volatilization on the test results during soil sample storage, the pot experiment was conducted in corresponding stages. After harvest, the fresh weight of the aboveground parts and the THg content of the leaves were measured.

3. Results

3.1. Soil biocurrent transport and influencing factors

The voltage and accumulated charge profiles of the MES filled with two kinds of soils (marked as A and B) were obviously different. For Soil A, the biocurrent was rapidly generated after the connected circuit on Day 3, whereas that of the MES filled with Soil B was observed on Day 35 (Fig. 1a). Furthermore, the voltage continuously increased during the 135-day operation for Soil A, with a maximum voltage of 21.7 ± 1.5 mV. In contrast, the maximum voltage of 6.8 ± 0.7 mV appeared on Day 87 for Soil B and then decreased gradually. Consequently, a cumulative charge of 130 ± 16 C was transferred in Soil A, which was 2.25-fold greater than that in Soil B. The power density of the soil MES significantly improved on Days 135 and 90 for Soils A and B, respectively (Fig. 1c), although the internal resistances of the soil were comparable in terms of the polarization curves (Fig. 1b and Fig. S2). In particular, for Soil A, the power density increased from 0.04 ± 0.002 on Day 90 to 0.08 ± 0.04 mW m^{-2} on Day 135, which was 267% greater than that for Soil

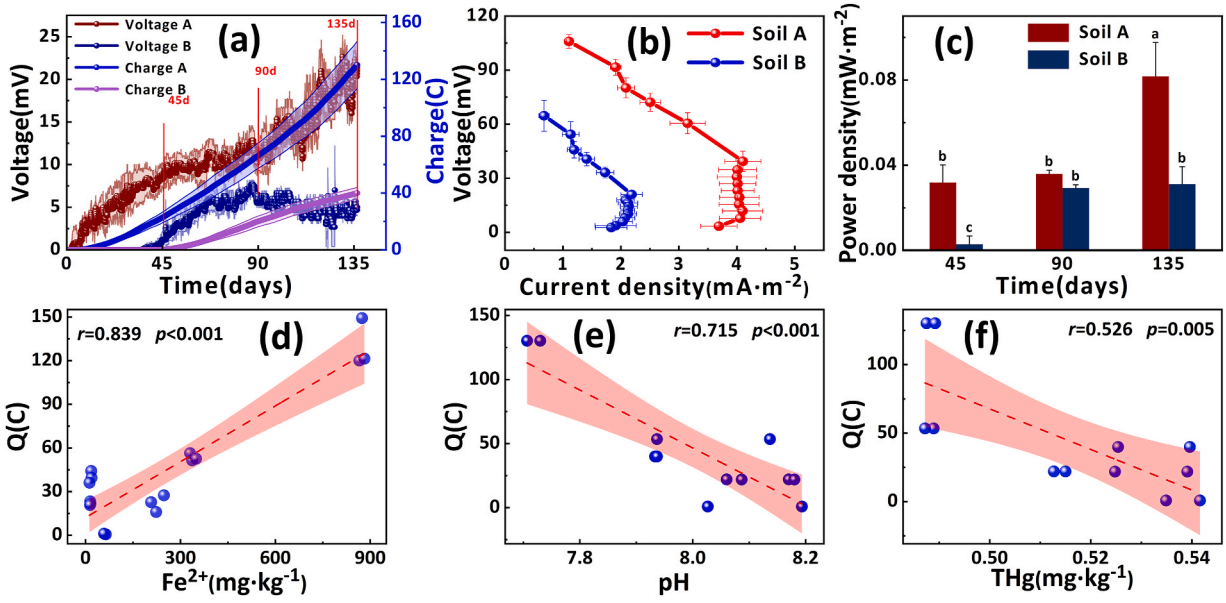


Fig. 1. Electrochemical properties of the soil MES with soils A and B. Voltage and accumulated charge (a), polarization curve on Day 135 (b), and maximum power density with time (c). Correlations between Fe(II), pH, THg and Q (d–f), with the dashed lines representing 95% confidence intervals. The error bars represent the standard error of the three repeated samples, and the different lowercase letters above the columns represent significant differences ($p < 0.05$). A and B represent two different types of soil.

B.

Through correlation analysis, it was found that the Fe(II) content, pH and THg content were the main factors affecting soil biocurrent transfer (Fig. 5a). Among them, Fe(II) was positively correlated with Q, and the linear correlation coefficient was 0.839 ($p < 0.001$, Fig. 1d). The pH and THg content were negatively correlated with Q, and the correlation coefficients were −0.715 ($p < 0.001$, Fig. 1e) and −0.526 ($p = 0.005$, Fig. 1f), respectively. In fact, the contents of TFe, Fe(III) and Fe(II) in Soil A were 24%, 164-fold, and 5-fold greater than those in Soil B (Table S1); thus, the change in Fe(III)/Fe(II) was also greater in the MES filled with Soil A, especially near the anode (Fig. S3). In contrast, the SO_4^{2-} content was more distinctly changed near the cathode of the MES filled with Soil A than in that filled with Soil B (Figs. S3e and 3f).

3.2. Effects of soil biocurrent on THg and Hg forms

The content of THg was significantly reduced in the soil MES. For Soil A, the THg content decreased by 5.5–7.1% and 4.6−7.6% in the cathodic and anodic soils of the closed-circuit group compared with the original soil (OS) on Days 90 and 135, respectively ($p < 0.05$, Fig. S4a). For Soil B, the lowest concentration of THg was observed in the anodic soil of the closed-circuit group on Day 135, with a decrease of 3.9% ($p > 0.05$, Fig. S4b). Compared with that of the corresponding OS, the THg content of the control in both soils did not change significantly at the end of the experiment, indicating that accelerating soil electron transfer could reduce the soil THg content. The output charge Q of the system exhibits a strong correlation with THg levels (Fig. 5a), suggesting a potential underlying relationship between biological current and THg content.

The soil biocurrent accelerated the aging of Hg forms in the early days. RS, the most stable form, accounted for 83−92% and 96−99% of the THg in Soils A and B, respectively, during the entire experimental period. In Soil A, the RS proportion in the cathodic and anodic soils of the closed-circuit group increased from 83% in the OS to 91% and 92%, respectively (Figs. S5b and 5d), whereas those in the corresponding control increased to 87% and 89% on Day 45, respectively (Figs. S5a and 5c). Similarly, the RS content in the Soil B closed-circuit system increased from 97% in the OS to 99% on Day 45 (Fig. S5). Interestingly, the decrease in OM was similar to the increase in RS, indicating that the Hg form transformed from OM to RS during the flooding treatment. During the whole experiment, the sum of EX, CB and OX accounted for less than 2% of THg. The decrease in the soil THg content was explained by the decrease in the RS content in the closed-circuit system after 45 days.

3.3. Effects of soil biocurrent on bio Hg and MeHg concentrations

After flooding, the concentration of Bio Hg in both soils first increased but then decreased. The Bio Hg contents of the Soil A anode and CK were 3.42 ± 0.85 and 4.08 ± 0.9 μg kg^{-1}, respectively, on Day 45 and 4.7 ± 0.9 and 6.7 ± 0.5 μg kg^{-1}, respectively, on Day 90. Compared with that in the CK treatment, the Bio Hg content in the anode of the Soil A closed-circuit decreased by 16% and 29% on Days 45 and 90, respectively (Fig. 2a). There was no significant change in Bio Hg content between the Soil A cathode and CK. For Soil B, there were no significant differences between the CK and closed-circuit treatments. The content of Bio Hg in Soil B was one order of magnitude lower than that in Soil A (Fig. 2b). Through correlation analysis, it was found that pH, Fe(III) and DOC were important factors affecting the content of Bio Hg (Fig. 5a). Among them, DOC was the most important influencing factor, and the correlation coefficient with Bio Hg was 0.321 ($p = 0.002$, Fig. 2g). The contributions of Fe(III) and pH to Bio Hg were low, with correlation coefficients of 0.210 ($p = 0.014$, Figs. 2f) and 0.145 ($p =$

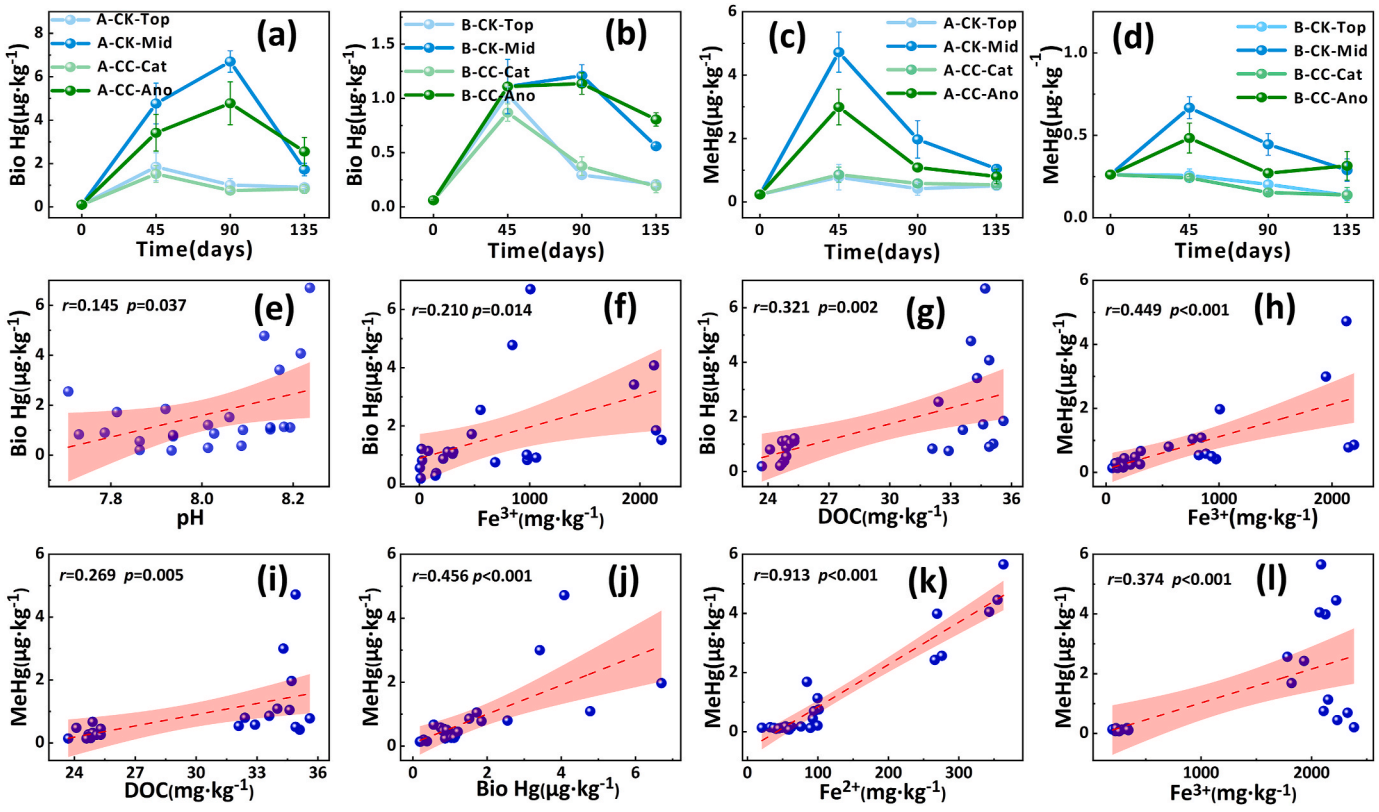


Fig. 2. Concentrations of Bio-Hg and MeHg in the two soils (a–d). Correlations between pH, Fe(II), DOC and Bio-Hg concentration (e–g) and between Fe(III), DOC, Bio-Hg and MeHg concentration (h–j) during the 135 days. Correlations between the Fe(II), Fe(III) and MeHg concentrations before Day 45 (k and i), with the dashed lines representing 95% confidence Intervals. CC-Cat and CC-Ano represent the closed-circuit cathodic and anodic regions, respectively. CK-Top and CK-Ano represent the corresponding sampling points of the control near the cathode and anode regions, respectively.

0.037, Fig. 2e), respectively. Overall, soil properties were the vital factors affecting Bio-Hg.

The trend of the MeHg content in the two soils was similar to that of Bio-Hg. The MeHg content was the highest on Day 45, and in the Soil A anode and corresponding CK, it was 4.7 ± 0.8 and 3.0 ± 0.9 μg kg^{-1}, respectively (Fig. 2c). The MeHg contents in the Soil B anode and corresponding CK were 0.7 ± 0.01 and 0.5 ± 0.01 μg kg^{-1}, respectively, on Day 45 (Fig. 2d). The MeHg content in the Soil A anode was 37% and 45% lower than that in CK on Days 45 and 90, respectively, and that in Soil B was 28% and 39%, respectively. There was no significant change in the MeHg content between the CK and the cathode of Soils A and B during the entire test period. Through correlation analysis, it was found that Fe(III), DOC and Bio Hg were important factors affecting the content of MeHg (Fig. 5a). The correlation coefficient between Bio-Hg and MeHg was 0.456 ($p < 0.001$, Fig. 2j), indicating that Bio Hg is a methylated Hg source. The correlation coefficients of Fe(III) and DOC with the MeHg content were 0.449 ($p < 0.001$, Figs. 2h) and 0.269 ($p = 0.005$, Fig. 2i), respectively, indicating that MeHg formation was typically determined by a dissimilatory iron reduction process dominated by IRB.

3.4. Effects of the soil biocurrent on the functional bacterial community

The community coverage for the treatments ranged from 98 to 99%, indicating that the gene sequencing depth can represent the situation of the microbial community (Table S3). According to the PCA results, the original Soils A and B plotted in the first quadrant, which was different from the other samples (Fig. S6), because flooding conditions changed the soil redox environment and affected the microbial composition. The samples from the cathode and corresponding CK plotted in the third quadrant, whereas the samples from the anode and corresponding CK mainly plotted in the fourth quadrant for Soils A and B. This difference was due to the different oxygen concentrations in the surface soil and subsoil. Consequently, the microbial richness and community diversity of the surface soils of the closed-circuit and control soils were greater than those of the subsoils. For example, for Soil A, the Chao1 index in the surface soil of the CK on Day 45 was 36% greater (2852 ± 89 vs. 2100 ± 230, respectively, $p < 0.05$, Table S3). In contrast, there was a slight difference between the closed circuit and the corresponding CK. Compared with those under OS, the relative abundances of *Proteobacteria*, *Acidobacteriota,* and *Gemmatimonadota* increased, whereas that of *Actinobacteria* decreased after flooding (Figs. S7a and 7b). For example, on Day 45, the abundance of *Proteobacteria* in the Soil A cathode and corresponding CK increased by 20–31%, and *Actinobacteria* decreased by 27–29%. The cluster heatmap of the dominant genera (top 30) revealed that the cathode and anode regions converged over time (Fig. S7c).

The dominant genera in the soil electroactive microorganisms were *Sphingomonas*, *Geobacter*, *Bacillus* and *Pseudomonas* (Fig. 3a and b). Except for the Soil A anode, the relative abundance of *Sphingomonas* was greater on Day 45 than on Day 135. For example, compared with that on Day 45, the relative abundance of *Sphingomonas* in the surface layer of the Soil A control and the Soil B cathode decreased by 49% and 52%, respectively, on Day 135. Interestingly, in Soil A, the relative abundance of *Geobacter* in the anode area and the corresponding CK area increased by 3.2- and 2.5-fold, respectively, on Day 45. However, the change in *Geobacter* abundance was negligible in Soil B, which corresponded to the profile of electricity generation.

Flooding and MES treatments altered the relative abundance of Hg methylation-related genera in the soil. The relative abundance of *Nitrospira* increased in the cathode and corresponding CK regions. For example, on Days 45 and 135, the abundance of *Nitrospira* in the surface layer of Soils A and B increased by 34–35% and 61−62%, respectively, relative to that in the OS (Fig. 3c and d). The relative abundance of *Geobacter* in Soil A clearly increased. On Day 45, the abundance of *Geobacter* in the anode and corresponding CK increased 42- and 9-fold, respectively. In Soil B, the relative abundance of *Clostridium* at the anode increased by 99% compared with that in OS on Day 45. By linear fitting between the abundance of Hg-methylated genera and the MeHg content, the correlation coefficients for *Nitrospira* and *Geobacter* were 0.058 ($p = 0.172$, Figs. 3e) and 0.494 ($p < 0.001$, Fig. 3f), respectively. This further suggested that *Geobacter* was responsible for MeHg conversion.

3.5. Effect of biocurrent on Hg conversion genes

Compared with OS, the soil MES increased the number of copies of the *hgcAB* gene in Soils A and B. For example, the number of copies of the *hgcAB* gene in the anodic and cathodic regions of Soil A increased by 801- and 499-fold, respectively, on Day 45 (Fig. 4a). Moreover, their copy numbers in the corresponding controls increased by 156- and 55-fold, respectively. In comparison, it increased by 5- and 9-fold in the cathode and anode, respectively, compared with the corresponding CK.

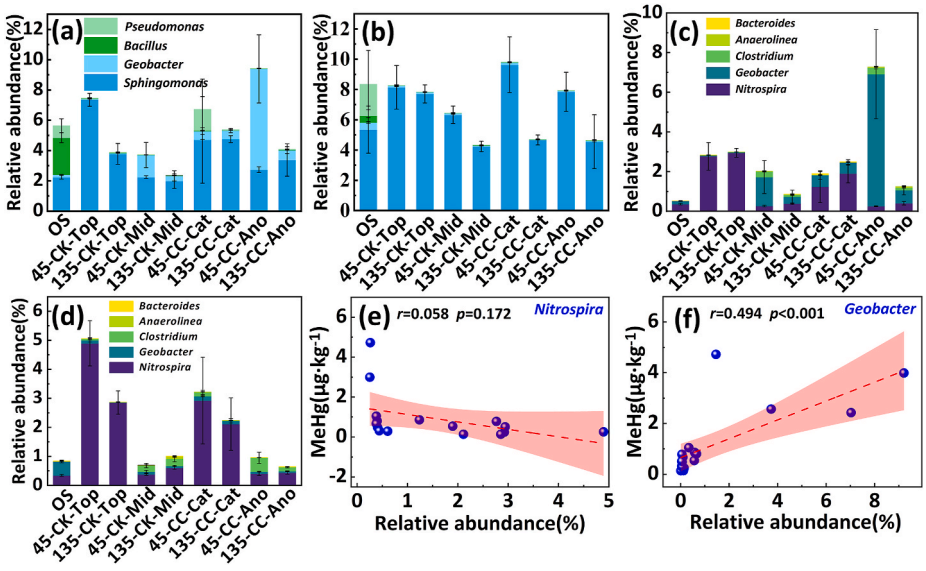


Fig. 3. The relative abundances of soil electroactive bacteria (top 30) in Soils A (a) and B (b). The relative abundance of soil Hg-methylating microorganisms (top 5) in Soils A (c) and B (d). Correlation analysis of *Nitrospira* (e) and *Geobacter* (f) abundance with MeHg content. The' 45-, 135- ' represents the time to obtain the sample. CC-Cat and CC-Ano represent the closed-circuit cathodic and anodic regions, respectively. CK-Top and CK-Ano represent the corresponding sampling points of the control near the cathode and anode regions, respectively.

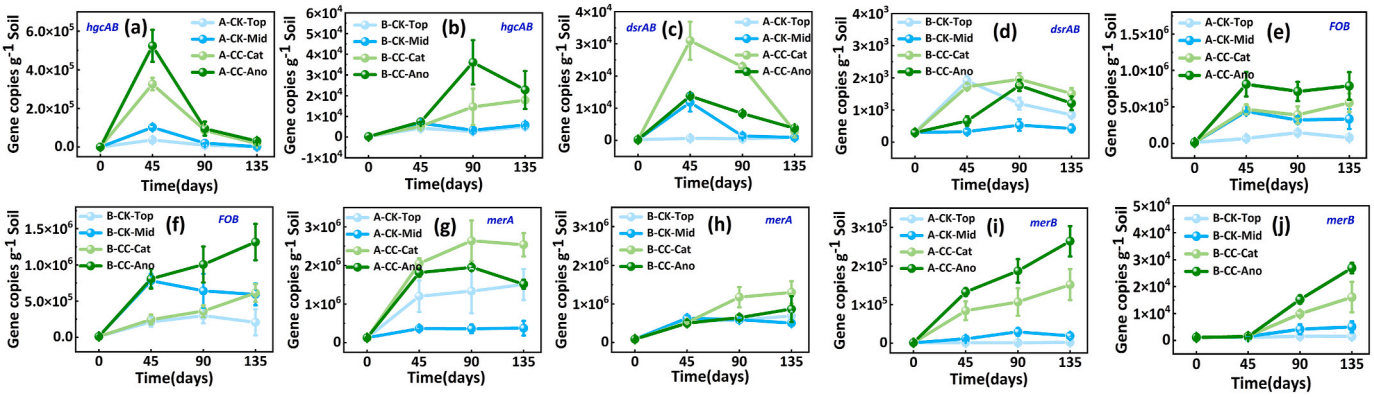


Fig. 4. Hg cycling functional gene copies in Soils A and B. Copies of *hgcAB* (a and b), *dsrAB* (c and d), *FOB* (e and f), *merA* (g and h), and *merB* (i and j). CC-Cat and CC-Ano represent the closed-circuit cathodic and anodic regions, respectively. CK-Top and CK-Ano represent the corresponding sampling points of the control near the cathode and anode regions, respectively.

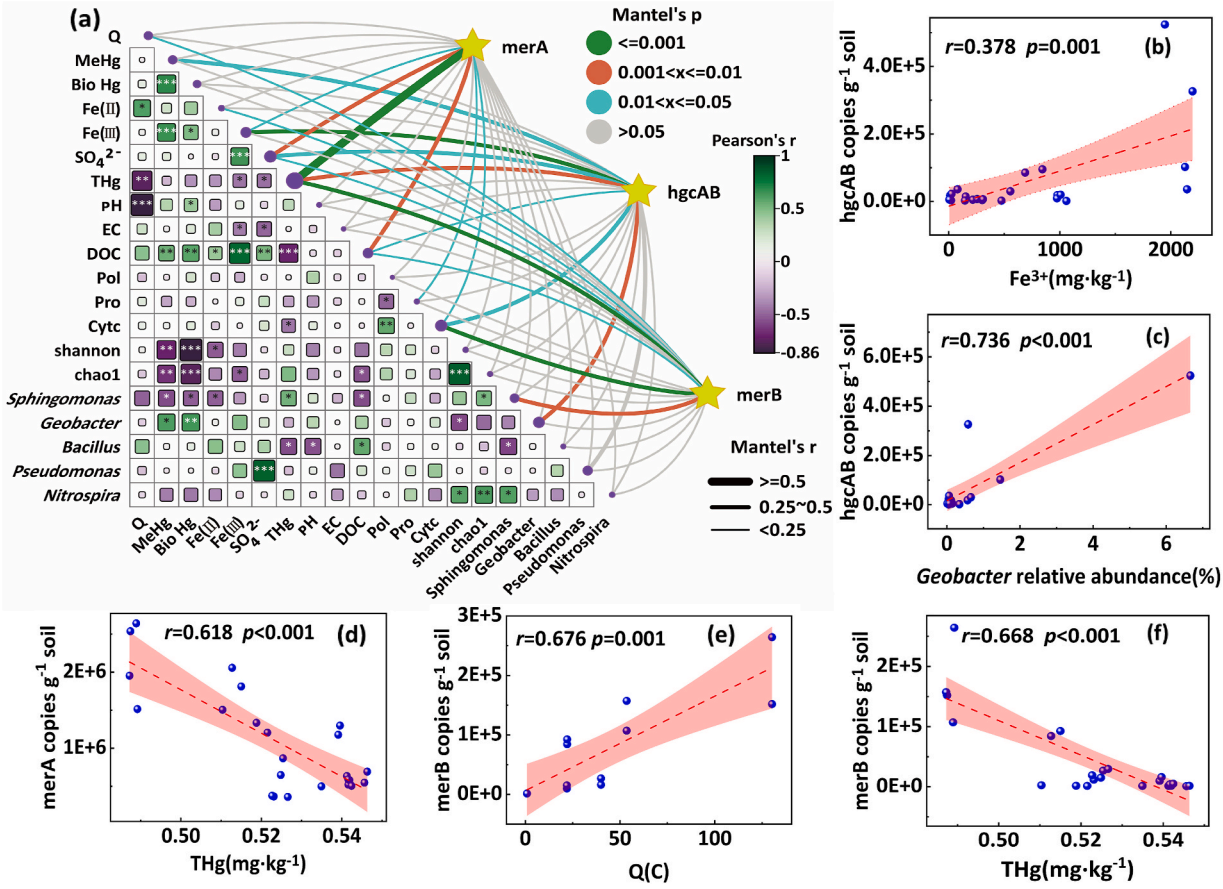


Fig. 5. Mantel test analysis of soil physicochemical properties, the bacterial α index and the abundance of Hg-methylated bacteria and Hg conversion genes. Correlation analysis of Fe(III) and *Geobacter* with *hgcAB* (b and c); THg with *merA* (d); and Q and THg with *merB* (f).

On Day 90, the number of copies of the *hgcAB* gene in Soil A decreased significantly compared with that on Day 45 ($p < 0.05$). For example, the number of copies of the *hgcAB* gene in the anode and cathode decreased by 82% and 74% and 84% and 74% in the corresponding CK, respectively. On Day 135, the number of copies of *hgcAB* in Soil A was comparable between the closed-circuit system and CK. The number of *hgcAB* copies in Soil B was 16–28 times greater than that in OS on Day 45, but the difference between the two treatments was not significant (Fig. 4b). The number of *hgcAB* copies in Soil B was the highest on Day 90, and the increase in the number of *hgcAB* copies in the anode and cathode was greater than that in the OS, with increases of 14- and 57-fold, respectively. On Day 135, its abundance in the anode decreased by 37% compared with that on Day 90, while it changed little over time in the CK. The increase in *hgcAB* gene copies in the anode and corresponding controls explained the increase in the MeHg content in the two soils.

The promotion effect of the soil MES on the number of *dsrAB* copies was greater than that of the control, which was mainly reflected in the cathode. The number of *dsrAB* in the cathode for Soil A was 118-fold greater than that in OS, whereas the number of *dsrAB* increased 53- and 45-fold in the anode and corresponding CK, respectively, on Day 45

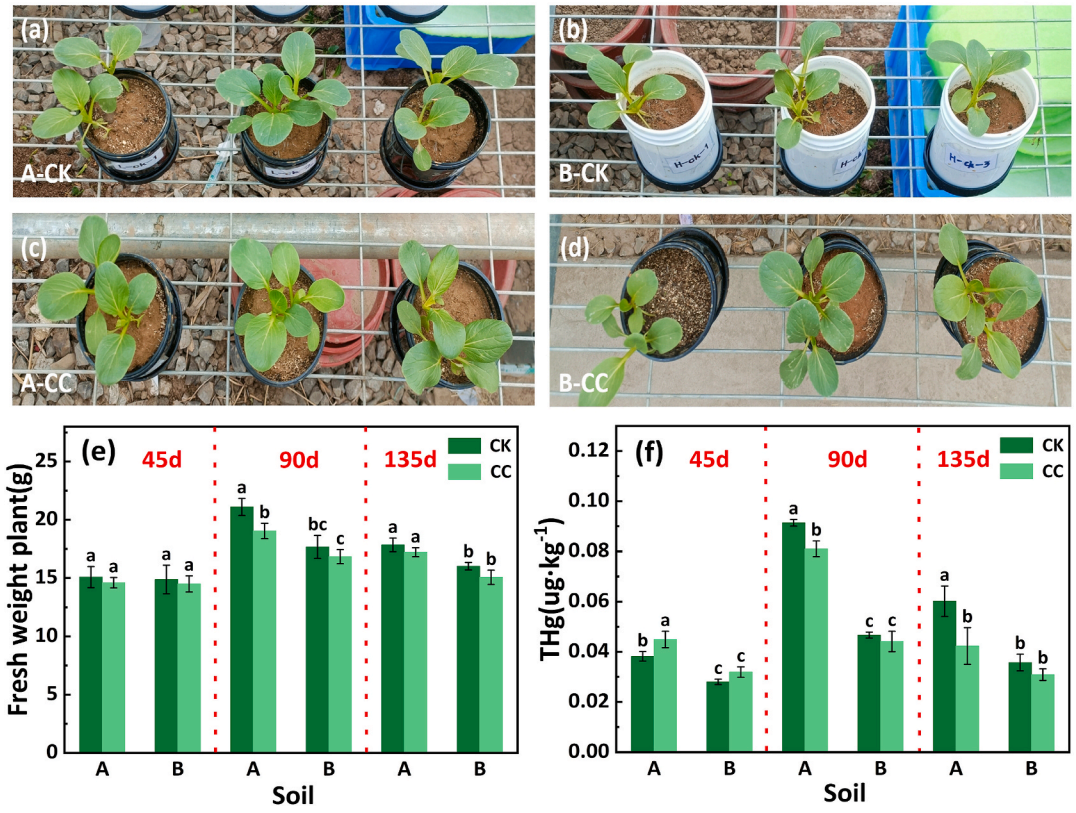


Fig. 6. Pot experiment with rape plants. The rape plant growth status of 135-day soils: Soil A CK (a) and closed-circuit (c) and Soil B CK (b) and closed-circuit (d). Weight of fresh plant (e) and THg content (f) of the upper part of the rape plant during the mature period. The different lowercase letters above the columns represent significant differences ($p < 0.05$).

(Fig. 4c). Sulfate reduction in Soil A was consistent with the change in *dsrAB* gene copy number (Fig. S3e). The change in the number of *dsrAB* copies in Soil B was similar to that in Soil A (Fig. 4d). Biocurrent promoted an increase in *FOB* gene copies, which increased more significantly in the anode than in the cathode. On Day 135, compared with those in OS, the number of *FOB* in the anode increased by 61- and 89-fold in the two soils, respectively (Fig. 4e and f). Compared with that in OS, in the two soils, the number of *FOB* in the cathode increased 43- and 65-fold, and the number of *FOB* in the subsoil in CK increased 26- and 46-fold on Day 135, respectively. The number of *FOB* in the two control surface soils was similar.

The biocurrent promoted an increase in the number of *merA* gene copies, which increased more significantly in the cathode than in the anode. On Day 135, compared with that on OS, the number of copies of *merA* in the cathode and anode increased by 19−15 and 12−10 times, respectively, in the two soils (Fig. 4g and h). Compared with that in OS, the number of copies of *merA* in the Soil A control increased by 3–11 times, whereas little change was detected between the other treatments in Soil B. The high abundance of the *merA* gene accelerated the atmospheric circulation of Hg in the soil, which potentially explained the decrease in the THg content in the soil. Under biological current stimulation, the number of copies of the *merB* gene increased significantly in the two soils ($p < 0.05$), and the anode effect was stronger than the cathode effect. On Day 135, compared with that in OS, the number of copies of *merB* in the anode and cathode increased by 23−181- and 14−104-fold, respectively, in the two soils (Fig. 4i and j). The number of copies of the *merB* gene did not significantly differ between the two control soils. The increase in *merB* gene copy number stimulated by the biocurrent accelerated MeHg reduction in the soil.

The Mantel test results revealed that the levels of MeHg, Fe(III), SO_4^{2-}, THg, Cytc and *Geobacter* were significantly correlated with the number of copies of the *hgcAB* gene ($p < 0.05$, Fig. 5a). The correlation coefficient between the MeHg content and *hgcAB* gene copy number was 0.228 ($p = 0.011$, Fig. S9a), indicating that Hg methylation was associated mainly with *hgcAB*-mediated biological processes. The correlation coefficients of Fe(III) and *Geobacter* with *hgcAB* were 0.378 ($p = 0.001$, Figs. 5b) and 0.736 ($p < 0.001$, Fig. 5c), respectively, indicating that Fe (III) and *Geobacter* were closely related to methylation. The SO_4^{2-}, THg, DOC, Pro and Cyt *c* contents were significantly correlated with the number of *merA* gene copies ($p < 0.05$), suggesting that the above factors had an important influence on the entry of Hg in the soil into atmospheric circulation. There was a significant negative correlation between THg and *merA*, with a correlation coefficient of 0.618 ($p < 0.001$, Fig. 5d). Q, Fe(II), SO_4^{2-}, THg, DOC, and Cyt *c* were significantly correlated with *merB* gene copies ($p < 0.05$), indicating that these factors play a vital role in MeHg demethylation. The correlation coefficients of Q with the *merB* gene were 0.676 ($p = 0.001$, Figs. 5e) and 0.668 ($p < 0.001$, Fig. 5f), respectively, and were negatively correlated with THg. This indicated that the increase in the *merB* gene caused by Q accelerated the reduction of MeHg to Hg(II) and decreased the content of MeHg. Moreover, Hg(II) was reduced to Hg^0 by *merA*, which reduced the content of THg.

3.6. Plant tests

The change of Bio Hg content in the whole soil MES was shown in Fig. S8. Overall, the closed-circuit system reduced Bio Hg content compared to CK. The growth of rape plants was similar between the Hg-contaminated soils in the MES and CK treatments (Fig. 6a–d). In the second batch of Soil A treated with the MES, the weight of fresh the upper part of the rape plant decreased by 9% compared with that of the CK ($p < 0.05$), but the weight of the upper part of the rape plant was similar in the other periods (Fig. 6e). In the first batch, the THg content of rape plants in Soil A treated with the MES increased by 13% compared

with that in the CK ($p < 0.05$, Fig. 6f). In the second and third batches, the THg content of rape plants was reduced by 11% and 30%, respectively, in treated Soil A compared with that in the CK ($p < 0.05$). The THg content of rape plants was lower in treated Soil B than in the CK, but the difference was insignificant. The plant pot experiment revealed that soil treated with the closed-circuit system not only had no adverse effect on the growth of rape, but also reduced Hg accumulation in the plants. This suggests that biocurrent treatment can decrease the bioavailability of Hg in the soil, highlighting its potential for broad application.

4. Discussion

The properties of the soil itself, along with the complex electron transfer network formed by microorganisms, determine the output performance of the soil MES (J. Zhang et al., 2023). In this study, we found that Fe(II), as an electron donor, directly influenced the power output between the two soil systems. Our analysis suggests that the minerals in red soil predominantly exist in a residual form, resulting in a very low content of active iron. From the speciation of Hg, we inferred that the mineral composition in soil B was largely responsible for this. More than 99% of the Hg in soil B remained in a residual state throughout the entire cycle. In addition, soil A contained a higher abundance of dominant electroactive microorganisms, and the increased availability of active iron facilitated the enrichment of *Geobacter* and *Sphingomonas*, while the dominant electroactive microorganisms in soil B were more limited, with *Sphingomonas* being the only prevalent species (Fig. 3). Furthermore, the higher levels of dissolved organic matter in soil A contributed to the superior output performance of the system. The greater availability of carbon sources for microbial growth and metabolism led to a more complex electron transfer network.

This study demonstrated that directionally accelerated electron transfer reduced the concentrations of THg, Bio Hg and MeHg in paddy soil and promoted the aging of Hg forms. There are two ways to age Hg. One is photoaging in which Hg(II) is mediated by light to generate mercuric sulfide nanoparticles (HgS_{NP}), and the other is the dark aging process; that is, Hg(II) is transferred from the weak binding site in the DOM molecule to the strong binding site to reduce its reduction ability (Shen et al., 2023). We speculate that the aging of Hg in the cathode region is caused by the formation of HgS_{NP}. Because the sulfate reduction process in the cathode area of soil A was stronger than that in the anode area throughout the entire test period (Fig. S3). In contrast, the aging of Hg in the subsoil may be regulated by the DOM-mediated dark aging process, thus reducing the effectiveness of Hg by generating a thiol–Hg complex (Shen et al., 2023).

Based on the experimental results, we propose the schematic diagram of the mechanism of biocurrent regulation of Hg transformation in paddy soil (Fig. 7a). There are three pathways for the soil MES to reduce Hg in paddy soil. (1) The competing electron between the anode and the mineral reduction led to the lack of electrons required for the reduction dissolution of Hg minerals (Pei et al., 2022; Pei et al., 2022), which was explained by the negative correlation between the THg content and electricity production ($p = 0.005$, Fig. 1f). Electron competition can influence both the efficiency of electron transfer and the rate of the mineral reduction process. For instance, when electron acceptors such as oxygen, nitrate, and Fe(III) are present, microorganisms tend to preferentially reduce those with higher energy yields, leading to a slower rate of Hg reduction (Shi et al., 2016). (2) The *merA* gene was upregulated to accelerate Hg(II) reduction to Hg^0 in the soil MES. In fact, the number of copies of the *merA* gene increased by 4–6-fold under biocurrent stimulation (Fig. 4g and h), which accelerated the consumption of Hg(II) and explained the decrease in THg in the MES. There was a positive correlation between Pro, Cytc, SO_4^{2-} and *merA* ($p < 0.05$, Fig. S9), indicating that they regulated the expression of the *merA* gene by affecting the charge output of the soil MES. (3) The soil MES accelerated Hg(II) consumption by upregulating the *hgcAB* gene and affecting Hg methylation, since a high level of the *hgcAB* gene usually indicates a high methylation rate (Zhou et al., 2022).

In the natural environment, Hg in soil is discharged by precipitation, mining exhaust, etc. (Li et al., 2024). In our experiment, the input of exogenous Hg was excluded. Therefore, in the absence of additional Hg sources, we observed a reduction in the effectiveness of Hg treatment during the later stages of the experiment, regardless of whether a closed-circuit or electrodeless treatment was used. Biocurrent enhanced the reduction of Hg(II) to Hg^0 by upregulating the expression of the *merA* gene, promoted the volatilization of Hg into the atmosphere, and subsequently reduced the THg content in the soil. In contrast, we did not observe a significant decrease in THg content in the non-MES treatment; therefore, Hg(II) was not reduced to Hg^0. In fact, the Bio Hg content decreased to a low level because of the aging of Hg, although the content of residue-state Hg accounted for >97% of the total Hg content in Soil B and >83% in Soil A. Consequently, the content of Bio Hg in Soils A and B differed by an order of magnitude, which was possibly due to the effect of organic matter in addition to Fe conversion, since the DOC content in Soil A was 33% higher than that in Soil B (Table S1), and a significant correlation was observed between the DOC and Bio Hg contents ($p = 0.002$, Fig. 2g). Previous studies have shown that soil properties (e.g., pH, redox potential, organic matter, iron and sulfur) affect the effectiveness of Hg (Sun et al., 2019; Wu et al., 2020). DOC serves as an energy and carbon source for microorganisms, stimulating the growth and metabolic activities of microbial communities, particularly those involved in Hg transformation, thereby increasing Hg bioavailability.

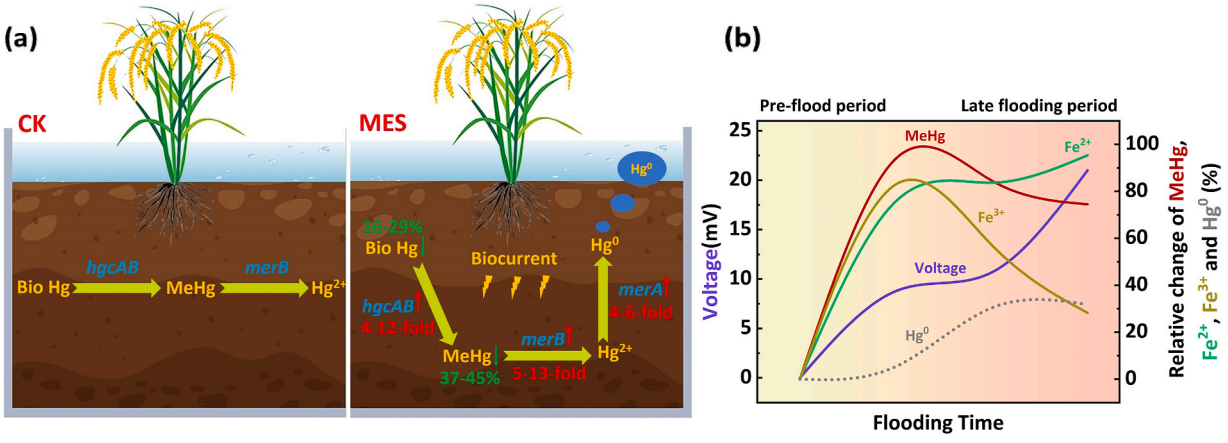


Fig. 7. Mechanistic diagram of Hg conversion in soil regulated by a biocurrent (a). Geochemical model of Hg conversion driven by a biocurrent in paddy soil (b). The content of Hg^0 is calculated from the reduction in THg.

Iron-reducing bacteria utilize organic carbon to reduce Fe(III) to Fe(II), and their metabolic processes may interact with Hg reduction or methylation pathways, influencing Hg bioavailability. The reduction of Fe(III) to Fe(II) creates a more favorable environment for Hg reduction, facilitating its conversion into a more bioavailable form (C. Chen et al., 2020).

In soil, redox potential and pH are important factors affecting the bioavailability of heavy metals (Meng et al., 2023). Flooding treatment leads to a decrease in the soil redox potential, resulting in an increase in metal mineral solubility (C. Chen et al., 2019; Zhou et al., 2022). The Bio Hg concentration in the surface soil did not fluctuate significantly over the study period, which may have been due to the oxygen content equilibrium in the air (X. Zhang et al., 2021). Our study revealed that pH was closely related to Bio Hg content ($p < 0.05$, Fig. 2e), and soil electron transfer led to a lower pH. Interestingly, previous studies have shown that higher pH promotes the formation of electroactive biofilms that accelerates electron transfer, but our results revealed that pH had a negative correlation with charge output ($p < 0.001$, Fig. 1e), indicating that soil acidification may also be conducive to the metabolic activity of electroactive microorganisms in certain soils (Yang et al., 2024; Yang et al., 2024). The pH and redox potential directly influence the mobility and bioavailability of Hg in soil by altering its chemical form, solubility, and microbial activity. Acidic and reducing conditions promote the dissolution and transformation of Hg into MeHg, thereby increasing its bioavailability. In contrast, alkaline and oxidizing conditions favor the precipitation of Hg, reducing its mobility and bioavailability.

Accelerated soil electron transfer stimulated the expression of the *mer* operon as a result of improved demethylation and reduced MeHg content ($p < 0.05$, Fig. 2). Although the methylation rate was greater than the demethylation rate in the subsoil before Day 45, the MeHg content ultimately distinctly decreased on Day 135. In contrast, there was an insignificant change in the MeHg content in the surface soil, which was determined by the low Bio Hg content and *hgcAB* gene. The rapid increase in gene copies (e.g., *hgcAB* and *dsrAB*) is not only a direct consequence of operating conditions, such as biological currents, but also closely linked to factors such as microbial growth kinetics, selection pressures, metabolic advantages, and horizontal gene transfer (Maddamsetti et al., 2024). Microorganisms can enhance their growth competitiveness and metabolic capacity by increasing the copies of specific functional genes in response to environmental changes, particularly in the context of heavy metal pollution, nutrient availability, and redox conditions, thereby improving their adaptability. Additionally, the number of copies of the *merB* gene increased 5–13-fold under biocurrent stimulation, which was supported by the significant correlation between the number of *merB* copies and the charge output ($p = 0.001$, Fig. 4e) and between the Fe(II) content and the Cyt *c* content ($p < 0.05$, Figs. S9f and 9g), indicating the formation of a biocurrent. As expected, Fe(II) was significantly correlated with charge output ($p < 0.05$) since Fe (II) shuttles through soil pores as an electron donor or directly mediates electron transfer (Lovley, 2022). Moreover, the addition of Fe minerals could promote an increase in the Fe(II) content and thus power generation (J. Zhang et al., 2023).

The up-regulation of these genes is usually related to the adaptive response of microorganisms to Hg-contaminated environments. For example, the *merA* gene encodes a Hg reductase that converts Hg^{2+} into Hg^0, which is a key step in Hg metabolism. The metabolic process of microorganisms requires electron donors and acceptors. The availability of electron donors in the environment can affect the Hg reduction capacity of microorganisms. In some cases, the up-regulation of *merA* gene may be related to the utilization of specific electron acceptors, especially when microorganisms are in oxygen-limited or anaerobic environments, the availability of electron acceptors will directly affect the efficiency of electron flow and Hg conversion. The electron transport chain (ETC) plays a key role in the metabolism of microorganisms, which transfers the electrons produced by metabolism to external electron acceptors. In the Hg reduction reaction involved in the *merA* gene, electrons are transferred from NADH or $FADH_2$ molecules produced by metabolism to Hg reductase. The efficiency and metabolic state of ETC directly affect the Hg reduction reaction (Z. Zhang et al., 2024). The presence and transformation of Hg not only influence microbial metabolic pathways but may also regulate the overall metabolic strategy of microorganisms through specific feedback mechanisms. For instance, during prolonged exposure to Hg-contaminated environments, microorganisms may enhance Hg conversion by increasing the copies of *merA/B* genes via gene amplification, mutation, or horizontal gene transfer (Hu et al., 2018). This amplification not only enables microorganisms to better manage Hg pollution but may also confer greater tolerance to other environmental stresses, such as heavy metals and oxidative stress. The Hg reduction process may interact with other metabolic pathways, such as the nitrogen and sulfur cycles. Microorganisms might modulate their nitrogen or sulfur metabolism pathways during Hg reduction to supply additional electron donors for the process. The interplay within this metabolic network enhances the adaptability of microorganisms, enabling them to effectively manage multiple stresses in complex environments (He et al., 2024).

The correlation between the amount of Fe(III) and *Geobacter* and the number of *hgcAB* copies ($p < 0.01$, Fig. 5b and c) indicated that *Geobacter* uses Fe(III) to complete the biotransformation of Hg through iron reduction. Furthermore, the abundance of *Geobacter* was correlated with the MeHg content ($p < 0.001$, Fig. 3), indicating that Fe reduction by *Geobacter* was the most important biomethylation process in this study. Biocurrent stimulated the enrichment of *Geobacter* and enhanced the iron reduction process in the soil. These microorganisms can reduce Hg^{2+} to Hg^0, a form more readily methylated through their metabolic activities, thereby increasing the potential for Hg methylation reactions. The synergistic interactions among microbial populations can effectively accelerate the rate of Hg methylation. Additionally, the biocurrent may promote Hg methylation by further fostering anaerobic conditions in the local environment. Interestingly, we found that SO_4^{2-} content was also correlated with the copies of the *hgcAB* gene through correlation analysis ($r = 0.245$, $p = 0.008$; Fig. S9b), although its contribution was lower than that of Fe(III) ($r = 0.378$, $p = 0.001$; Fig. 5b). This suggests that SRB may play a role second only to IRB in the formation of MeHg. This finding is also consistent with previous studies, which indicate that microbial methylation is a complex, networked coupling process (Wu et al., 2020; L. Zhang et al., 2019). The hydroxyl and superoxide radicals produced by the dark oxidation of Fe can directly degrade MeHg to Hg^0, which is an abiotic demethylation process (Xie et al., 2024). Therefore, MeHg demethylation in Fe-rich soil may be the combined effect of bioprocesses and dark Fe oxidation. Future research could focus on the following directions to further explore the interaction between biological and abiotic demethylation processes. Mechanisms of bio-mineral interactions: conducting in-depth studies on the interactions between microorganisms and mineral surfaces, with a focus on how microbial metabolites interact with minerals to promote the conversion of MeHg. Influencing factors of abiotic demethylation: investigating how abiotic factors, such as free radical generation, light, and the redox environment, collectively influence the demethylation process of MeHg in the context of microbial activity. Synergistic effects of environmental factors: using controlled experiments and environmental monitoring to examine the impact of environmental conditions (such as pH, temperature, and oxygen concentration) on both biological and abiotic demethylation processes, and exploring how these factors can be optimized to enhance MeHg degradation.

Finally, we propose a geochemical model of Hg conversion driven by the soil biocurrent (Fig. 7b). Flooding resulted in a decrease in redox potential, which facilitated the reduction of minerals and promoted the enrichment of anaerobic microorganisms. The constructed soil closed-circuit system enhanced the iron reduction process, promoted the enrichment of electroactive microorganisms such as *Geobacter*, and improved the power output of the system. As an initiating factor, the biocurrent activates the soil system and promotes the upregulation of

gene expression. Under biocurrent stimulation, the level of the *merA/B* gene increased, which promoted the demethylation of MeHg, accelerated the reduction of Hg(II) to Hg^0, and ultimately reduced the content of MeHg and THg. In this preliminary model, Fe(II) shows a strong correlation with the biocurrent, while Fe(III) exhibits a strong correlation with MeHg. We found that the Fe(III) content can predict changes in MeHg in soil on a long-term scale. Fe(III) was significantly correlated with MeHg within 135 days ($p < 0.001$), whereas Fe(II) was not correlated ($p = 0.227$, Fig. S9h). A recent study revealed that a decrease in the MeHg content was significantly correlated with a decrease in the Fe(II) content (Zhong et al., 2024). Therefore, we performed linear fitting of HCl-extracted Fe and MeHg contents in the early stage (before 45 days). The results showed that Fe(II) ($r = 0.913$, $p < 0.001$; Fig. 2k) had a stronger predictive effect on MeHg in the short term than did Fe(III) ($r = 0.374$, $p < 0.001$; Fig. 2l). Therefore, we hypothesized that Fe directly acts as a metabolic substrate of IRB to affect the activity and community structure of Hg-methylating microorganisms. Fe(II) produced through Fe(III) reduction acts as an electron donor for methylating microorganisms, facilitating the methylation process (T. Zhang et al., 2014). In an anoxic environment, certain methylating microorganisms can utilize Fe(II) to convert Hg into MeHg via methylation. Fe(II) serves not only as an electron donor for this process but also as a cofactor for the methylase reaction. Fe(III)-reducing bacteria influence the chemical speciation of Hg through electron transfer processes (J. Wang et al., 2024). Upon reduction of Fe(III) to Fe(II), Fe(II) can directly participate in Hg reduction and also impact the mobility and bioavailability of Hg in soil or sediment (Wang et al., 2016). For example, Fe(II) can form complexes with Hg, enhancing its solubility or altering its chemical state, which in turn affects the formation of MeHg (Xiang et al., 2022). With the involvement of IRB, microbial communities in soil or sediment may collaboratively promote both Fe(III) reduction and Hg transformation through cooperative mechanisms. For instance, SRB and methylating bacteria may form symbiotic relationships with Fe(III)-reducing bacteria, utilizing Fe(II)- or Fe(III)-mediated reactions to enhance MeHg production. Our research is crucial for understanding the biogeochemical pathways of Fe and its influence on Hg transformation, as well as for assessing and managing the ecological risks associated with Hg pollution.

For the large-scale application, improving the soil electrical conduction (e.g., conductive fulvic acid) and addition of electron donors (e.g., Fe^{2+}) are the potential ways for the soil microbial electrochemistry method. Therein, it is important to consider the effects of soil type and properties, and Hg concentration and availability as well as electron transfer capacity on system performance, while conducting a cost-benefit analysis. Furthermore, the long-term ecological risk assessment is crucial to ensuring the sustainability of method.

5. Conclusions

This study revealed that accelerated electron transfer could decrease the contents of Bio Hg, MeHg and THg in soil and verified the safety of the soil MES via plant experiments. On the one hand, the soil MES accelerates the reduction of Hg(II) in soil to Hg^0 in atmospheric circulation by upregulating the *merA* gene and decreasing THg in soil. On the other hand, by stimulating an increase in the *merB* gene, the reduction in and demethylation of MeHg were accelerated. Our research could potentially apply in Hg methylation hotspots such as paddy soils, wetlands and sediments. The ability of the soil MES to safely remediate Hg-contaminated paddy soil is highly important for the realization of the United Nations Environment Programme (UNEP) Minamata Convention on Hg.

CRediT authorship contribution statement

Yun Zhang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Yuewei Yang:** Methodology, Formal analysis, Data curation. **Xin Yu:** Resources, Formal analysis, Data curation. **Side Yang:** Software, Resources, Methodology. **Jialu Sun:** Software, Resources, Formal analysis. **Tao Sun:** Resources, Methodology, Investigation. **Xiaolin Zhang:** Validation, Software, Resources. **Huike Ye:** Software, Formal analysis. **Yuebing Sun:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation. **Xiaojing Li:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2025.144657.

Data availability

Data will be made available on request.

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